

```

> //Element query operator
>
> db.products.find(
  {
    price: {
      $exists: true
    }
  },
  {
    name: 1,
    price: 1
  }
)
< {
  _id: 1,
  name: 'xPhone',
  price: 799
}
{
  _id: 2,
  name: 'xTablet',
  price: 899
}
{
  _id: 3,
  name: 'SmartTablet',
  price: 899
}
{
  _id: 4,
  name: 'SmartPad',
  price: 699
}

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{
  _id: 5,
  name: 'SmartPhone',
  price: 599
}
> db.products.find({
  price: {
    $exists: true,
    $gt: 699
  }
}, {
  name: 1,
  price: 1
});
< {
  _id: 1,
  name: 'xPhone',
  price: 799
}
{
  _id: 2,
  name: 'xTablet',
  price: 899
}
{
  _id: 3,
  name: 'SmartTablet',
  price: 899
}
> db.products.find({
  price: {
    $exists: false
  }
}

```

```
}, {
  name: 1,
  price: 1
});
<
> db.products.find({
  price: {
    $type: "string"
  }
}, {
  name: 1,
  price: 1
})
<
> db.products.find({
  price: {
    $type: "number"
  }
}, {
  name: 1,
  price: 1
})
< {
  _id: 1,
  name: 'xPhone',
  price: 799
}
{
  _id: 2,
  name: 'xTablet',
  price: 899
}
{
  _id: 3,
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    price: 899
  }
  {
    _id: 4,
    name: 'SmartPad',
    price: 699
  }
  {
    _id: 5,
    name: 'SmartPhone',
    price: 599
  }
}
> db.products.find({
  price: {
    $type: "array"
  }
}, {
  name: 1,
  price: 1
})
<
> db.products.find({
  price: {
    $type: ["number", "string"]
  }
}, {
  name: 1,
  price: 1
})
< {
  _id: 1,
  name: 'xPhone',
  price: 799

```

```

{
  _id: 2,
  name: 'xTablet',
  price: 899
}
{
  _id: 3,
  name: 'SmartTablet',
  price: 899
}
{
  _id: 4,
  name: 'SmartPad',
  price: 699
}
{
  _id: 5,
  name: 'SmartPhone',
  price: 599
}
}
> db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate" : ISODate("2011-05-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" : 2.6
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate" : ISODate("2011-09-01T00:00:00Z"), "spec" : { "ram" : 16, "screen" : 9.5, "cpu" : 3
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate" : ISODate("2015-01-14T00:00:00Z"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu"
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate" : ISODate("2020-05-14T00:00:00Z"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" : 1
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate" : ISODate("2022-09-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 9.7, "cpu" :
  { "_id" : 6, "name" : "xWidget", "spec" : { "ram" : 64, "screen" : 9.7, "cpu" : 3.66 }, "color" : [ "black" ], "storage" : [ 1024 ] }
  ])
MongoBulkWriteError: E11000 duplicate key error collection: test.products index: _id_dup key: { _id: 1 }
> db.products.find({
  color: {
    $size: 2

```

```

    }
  }, {
    name: 1,
    color: 1
  })
< {
  _id: 1,
  name: 'xPhone',
  color: [
    'white',
    'black'
  ]
}
> db.products.find({
  $or: [{
    color: {
      $size: 1
    }
  },
  {
    color: {
      $size: 2
    }
  }
  ]
}, {
  name: 1,
  color: 1
})

```

```

  _id: 1,
  name: 'xPhone',
  color: [
    'white',
    'black'
  ]
}
{
  _id: 3,
  name: 'SmartTablet',
  color: [
    'blue'
  ]
}
> db.products.find({
  color: {
    $all: ["black", "white"]
  }
}, {
  name: 1,
  color: 1
})
< {
  _id: 1,
  name: 'xPhone',
  color: [
    'white',
    'black'
  ]
}
{
  _id: 2,
  name: 'xTablet',

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    color: [
      'white',
      'black',
      'purple'
    ]
  }
}
> db.products.find({
  $and: [
    {color: "black"},
    {color: "white"}
  ]
}, {
  name: 1,
  color: 1
})
< {
  _id: 1,
  name: 'xPhone',
  color: [
    'white',
    'black'
  ]
}
{
  _id: 2,
  name: 'xTablet',
  color: [
    'white',
    'black',
    'purple'
  ]
}

```

```

> db.products.find({
  storage: {
    $elemMatch: {
      $lt: 128
    }
  }
}, {
  name: 1,
  storage: 1
});
< {
  _id: 1,
  name: 'xPhone',
  storage: [
    64,
    128,
    256
  ]
}
{
  _id: 3,
  name: 'SmartTablet',
  storage: [
    16,
    64,
    128
  ]
}
> db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate" : ISODate("2011-05-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" : 2

```

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    })
    ○ • MongoBulkWriteError: E11000 duplicate key error collection: test.products index: _id_ dup key: { _id: 1 }
    > db.products.find({
      'price': {
        $exists: 1
      }
    }, {
      name: 1,
      price: 1
    }).sort({
      price: 1
    })
    < {
      _id: 5,
      name: 'SmartPhone',
      price: 599
    }
    {
      _id: 4,
      name: 'SmartPad',
      price: 699
    }
    {
      _id: 1,
      name: 'xPhone',
      price: 799
    }
    {
      _id: 2,
      name: 'xTablet',
      price: 899
    }
  ,

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```

    {
      _id: 3,
      name: 'SmartTablet',
      price: 899
    }
    > db.products.find({
      'price': {
        $exists: 1
      }
    }, {
      name: 1,
      price: 1
    }).sort({
      price: 1
    })
    < {
      _id: 5,
      name: 'SmartPhone',
      price: 599
    }
    {
      _id: 4,
      name: 'SmartPad',
      price: 699
    }
    {
      _id: 1,
      name: 'xPhone',
      price: 799
    }
    {
      _id: 2,
      name: 'xTablet',

```

```
    price: 1
  })
< {
  _id: 5,
  name: 'SmartPhone',
  price: 599
}
{
  _id: 4,
  name: 'SmartPad',
  price: 699
}
{
  _id: 1,
  name: 'xPhone',
  price: 799
}
{
  _id: 2,
  name: 'xTablet',
  price: 899
}
{
  _id: 3,
  name: 'SmartTablet',
  price: 899
}
test>
```